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Theory assignment -1

Q1)- Define an algorithm. List any four characteristics of an algorithm.

→ An algorithm is a procedure used for solving a problem or performing a computation

The four characteristics of algorithm is as follows:

(i) An algorithm must have well-defined inputs.

(ii) It should produce well-defined outputs.

(iii) The algorithm ~~must~~ ~~have~~ ~~well~~ should be infinite, meaning it must terminate after a limited number of steps.

(iv) All steps must be feasible and practically executable.

Q2)- Define data structure and Abstract Data Type (ADT). Give one example each.

→ A Data structure is a way of organising information so that it is easier to use

The example of Data structure is Array, Linked List etc.

A Abstract Data Type (ADT) is a programming concept that defines a high level view of the data structure without specifying the implementation detail

The example of abstract Data Type is List, Array, stack, Queue etc.

Q3)- Explain the process of converting a problem into a program. Illustrate the role of algorithms in this process.

→ The process of converting a problem ~~and categorize~~ into ~~which type of problem it is~~: a program is as follows:

(i) First we identify the problem & categorize which type of problem it is.

(ii) Then analyse the problem.

(iii) In third step, we develop a algorithm, we develop a algorithm which helps us to solve the problem steps-by-steps.

(iv) In fourth step, we write a program for the problem.

(v) At last we test the program that we had written for the problem.

The role of Algorithm in this process is as follows:

(i) Algorithm are necessary for solving complex problem efficiently and effectively

(ii) They help to automate process and make them more reliable, faster and easier to perform.

Q4)- Explain linear and non-linear data structures with suitable example.

→ • A linear data structure traverses the data elements sequentially, in which only one data elements can directly be reached.

Ex: Arrays, Linked Lists, Stacks, Queues.

• Every data item is attached to several other data items in a way that is specific for reflecting relationships. The data items are not arranged in a sequential structure.

Ex: Trees, Graphs, Heaps.

Q5)- Differentiate between algorithm and program. Analyze their roles in the Software Development Life Cycle (SDLC).

→

ALGORITHM	PROGRAM
(i) Algorithm is a step by step procedure to solve a program	(i) A set of instructions written in a specific programming language

(ii) Abstract & Theoretical	(ii) Concrete & practical
(iii) Language Independent	(iii) ~~Concrete & practical~~ Language dependent
(iv) Cannot be executed by computer	(iv) Can be executed by a computer.

The phases of SDLC is as follows:

SDLC
↓
Requirement Analysis
↓
Design
↓
Implementation
↓
Testing
↓
Evolution

The role of algorithm is to help simulate or model project feasibility and estimate resources, timeline and costs. They also ensure correctness and performance of software and generate test cases for validation. It also ensures compatibility, load balancing and efficient deployment.

Q6)- Examine the relationship among, data structure, and algorithm. Justify how the choice of data structure affects algorithm efficiency.

→ Data structures and algorithms are two interrelated concepts in computer science. Data structures refer to the organization, storage, and retrival of data, while algorithms refer to the set of instructions used to solve a particular problem or perform a specific task.

Data structure and algorithm are often used together to create efficient and effective software solutions. For example, a sorting algorithm can be used to sort data in a array while a search algorithm can be used to sort data in a array, while a search algorithm can be used to find a particular item in a linked list.

The choice of data structure and algorithm depends on the type of data being processed and the nature of the problem to be solved. By selecting the most appropriate data structure and algorithm, software developers can create high performance solutions that are both efficient and effective.